



Sec: OUTGOING(SC-R,N-CHAINA, N120,CO-SPARK,SPARK)

Dt: 03-07-21

Time: 3 Hrs

SGTM-4

Max.Maks: 300

PHYSICS

Max.Marks: 100

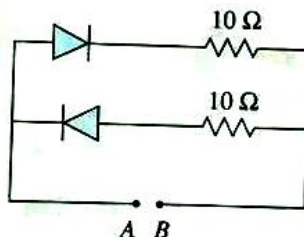
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

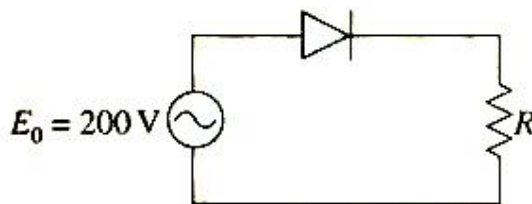
1. Ultraviolet light of wavelength 300 nm and intensity 1.0 W m^{-2} falls on the surface of a photosensitive material. If one percent of the incident photons produce photoelectrons, then the number of photoelectrons emitted per second from an area of 1.0 cm^2 of the surface is nearly
A) $9.61 \times 10^{14} \text{ s}^{-1}$ B) $4.12 \times 10^{13} \text{ s}^{-1}$ C) $1.51 \times 10^{12} \text{ s}^{-1}$ D) $2.13 \times 10^{11} \text{ s}^{-1}$
2. A surface irradiated with light $\lambda = 480 \text{ nm}$ gives out electrons with maximum velocity $v \text{ ms}^{-1}$, the cut off wavelength being 600 nm. The same surface would release electrons with maximum velocity $2v \text{ ms}^{-1}$ if it is irradiated by light of wavelength.
A) 325 nm B) 360 nm C) 384 nm D) 300 nm
3. The photoelectric threshold of a certain metal is $3000 \text{ \AA}$. If the radiation of $2000 \text{ \AA}$ is incident on the metal
A) electrons will be emitted B) positrons will be emitted
C) protons will be emitted D) electrons will not be emitted
4. If 5% of the energy supplied to a bulb is irradiated as visible light, how many quanta are emitted per second by a 100 W lamp? Assume wavelength of visible light as $5.6 \times 10^{-5} \text{ cm}$.
A) 1.4×10^{19} B) 3×10^3 C) 1.4×10^{-19} D) 3×10^4
5. An electron of mass m_e and a proton of mass m_p are accelerated through the same potential difference. The ratio of the de Broglie wavelength associated with an electron to that associated with proton is
A) 1 B) m_p / m_e C) m_e / m_p D) $\sqrt{m_p / m_e}$
6. What is the de Broglie wavelength of the wave associated with an electron that has been accelerated through a potential difference of 50.0 V?
A) 2.7×10^{-10} B) 1.74×10^{-10} C) 3.6×10^{-9} D) 4.9×10^{-11}
7. Up to what potential V can a zinc ball (work function 3.74 eV) removed from other bodies be charged by irradiating it with light of $\lambda = 200 \text{ nm}$?
A) 2.5 V B) 1.8 V C) 2.2 V D) 3 V
8. The human eye is most sensitive to green light of wavelength 505 nm. Experiments have found that when people are kept in a dark room until their eyes adapt to the darkness, a single photon of green light will trigger receptor cells in the rods of the retina. The velocity of typical bacterium of mass $9.5 \times 10^{-12} \text{ g}$, if it had absorbed all energy of photon, is nearly
A) 10^{-6} ms^{-1} B) 10^{-8} ms^{-1} C) 10^{-10} ms^{-1} D) 10^{-13} ms^{-1}

9. A photon collides with a stationary hydrogen atom in the ground state inelastically. Energy of the colliding photon is 10.2 eV. Almost instantaneously, another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector ?
 A) Two photons of energy 10.2 eV B) Two photons of energy 1.4 eV
 C) One photon of energy 10.2 eV and one electron of energy 1.4 eV
 D) One electron having kinetic energy nearly 11.6 eV
10. The voltage applied to an X-ray tube is 18 kV. The maximum mass of photon emitted by the X-ray tube will be
 A) 2×10^{-13} kg B) 3.2×10^{-36} kg C) 3.2×10^{-32} kg D) 9.1×10^{-31} kg
11. When the voltage applied to an X-ray tube increases from $V_1 = 10$ kV to $V_2 = 20$ kV, the wavelength interval between K_α line and cut-off wavelength of continuous spectrum increases by a factor of 3. Atomic number of the metallic target is
 A) 28 B) 29 C) 65 D) 66
12. Monochromatic radiations of wavelength λ are incident on a hydrogen sample in the ground state. Hydrogen atom absorbs the light and subsequently emits radiations of 10 different wavelengths. The value of λ is nearly
 A) 203 nm B) 95 nm C) 80 nm D) 73 nm
13. The power of an X-ray tube is 16 W. If the potential difference applied across the tube is 5 kV, then the number of electrons striking the target per second is
 A) 8.4×10^{16} B) 5×10^{17} C) 2×10^{16} D) 2×10^{19}
14. Assuming that about 200 MeV of energy is released per fission of ${}_{92}\text{U}^{235}$ nuclei, the mass of U^{235} consumed per day in a fission reactor of power 1 megawatt will be approximately
 A) 10^{-2} g B) 1g C) 100g D) 10,000g
15. If mass of $\text{U}^{235} = 235.12142$ amu, mass of $\text{U}^{236} = 236.1205$ amu, and mass of neutron = 1.008665 amu, then the energy required to remove one neutron from the nucleus of U^{236} is nearly about
 A) 75 MeV B) 6.5 MeV C) 1 eV D) Zero
16. A diode having potential difference 0.5 V across its junction which does not depend on current is connected in series with resistance of 20Ω across source. If 0.1 A pass through resistance, then what is the voltage of the source ?
 A) 1.5 V B) 2.0 V C) 2.5 V D) 5 V
17. A 2V battery is connected across the points A and B as shown in the figure given below. Assuming that the resistance of each diode is zero in forward bias and infinity in reverse bias, the current supplied by the battery when its positive terminal is connected to A is



- A) 0.2 A B) 0.4 A C) Zero D) 0.1 A

18. A sinusoidal voltage of peak value 200 V is connected to a diode and resistor R in the circuit shown so that half wave rectification occurs. If the forward resistance of the diode is negligible compared to R, the rms voltage (in volt) across R is approximately



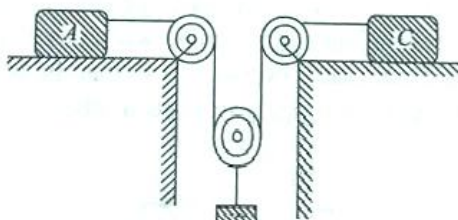
- A) 200 B) 100 C) $\frac{200}{\sqrt{2}}$ D) 280
19. In an FM system a 7 kHz signal modulates 108 MHz carrier so that frequency deviation is 50 kHz. The carrier swing is
A) 7.143 B) 8 C) 0.71 D) 350
20. A transmitter transmits a power of 10 kW when modulation is 50%. Power of carrier wave is
A) 5 kW B) 8.89 kW C) 14 kW D) 5.7 kW

SECTION – II (NUMERICAL VALUE ANSWER TYPE)

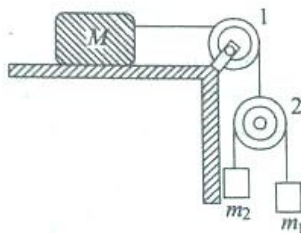
This section contains 10 questions. The answer to each question is a Numerical value. If the numerical value has more than two decimal places, **Truncate/Round-off the value** of Two decimal places. **Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:
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21. A 100 W light bulb is placed at the centre of a spherical chamber of radius 20 cm. Assume that 60% of the energy supplied to the bulb is converted into light and that the surface of the chamber is perfectly absorbing. Find the pressure exerted by the light on the surface of the chamber ($\text{___} \times 10^{-7} \text{ Pa}$).
22. A silver ball of radius 4.8 cm is suspended by a thread in a vacuum chamber. Ultraviolet light of wavelength 200 nm is incident on the ball for some time during which a total light energy of $1.0 \times 10^{-7} \text{ J}$ falls on the surface. Assuming that on the average one photon out of 10,000 is able to eject a photoelectron, find the electric potential ($\text{___} \times 10^{-1} \text{ V}$) at the surface of the ball assuming zero potential at the surface of the ball assuming zero potential at infinity.
23. A light beam of wavelength 400 nm is incident on a metal plate of work function 2.2 eV. A particular electron absorbs a photon and makes two collisions before coming out of the metal. Assuming that 10% of the extra energy is lost to the metal in each collision, find the maximum number of collisions the electron can suffer before it is unable to come out of the metal.

24. A hydrogen like atom (atomic number Z) is in a higher excited state of quantum number n . The excited atom can make a transition to the first excited state by successively emitting two photons of energy 10.2 eV and 17.0 eV respectively. Alternatively, emitting two photons of energies 4.25 eV and 5.95 eV, respectively. Determine the values of Z . (Ionization energy of H – atom is 13.6 eV).
25. The electric potential between a proton and an electron is given by, $V = V_0 \left(\frac{r}{r_0} \right)$ where V_0 and r_0 are constants and r is the radius of the electron orbit around the proton. Assuming Bohr's model to be applicable, it is found that r is proportional to n^x , where n is the principal quantum number. Find the value of x .
26. An ambulance sounding a horn of frequency 256 Hz is moving towards a vertical wall with a velocity of 5 m/sec. If the speed of the sound is 330 m/sec, how many beats per second will be heard by an observer standing a few metres behind the ambulance?
27. The sound level at a point 5.0 m away from a point source is 40 dB. What will be the level ($\text{---} \times 10$ dB) at a point 50 m away from the source?
28. A bullet passes past a person at a speed of 220 m/s. Find the fractional change in the frequency of the whistling sound ($\text{---} \times 10^{-1}$) heard by the person as the bullet crosses the person. Speed of sound in air is 330 m/s.
29. Blocks A and C start from rest and move to the right with acceleration $a_A = 6t \text{ m/sec}^2$ and $a_C = 3t \text{ m/sec}^2$. Here t is in seconds. The time when block B again comes to rest in W sec. Find W .



30. In the arrangement shown in figure, $m_1 = 1 \text{ kg}$ and $m_2 = 2 \text{ kg}$. The pulleys are massless and strings are light. The value of M for which the mass m_1 moves with constant velocity (neglecting friction) is W kg. Find W .



CHEMISTRY

Max.Marks:100

SECTION - I
(SINGLE CORRECT ANSWER TYPE)

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31. The correct relationships between standard free energy change in a reaction and the corresponding equilibrium constant K_c is

A) $\Delta G^0 = RT \ln K_c$

B) $-\Delta G^0 = RT \ln K_c$

C) $\Delta G = RT \ln K_c$

D) $-\Delta G = RT \ln K_c$

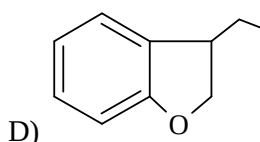
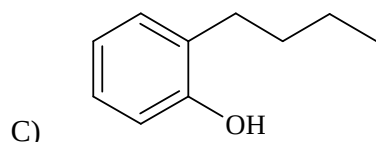
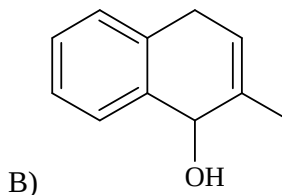
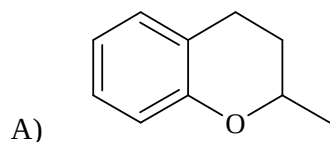
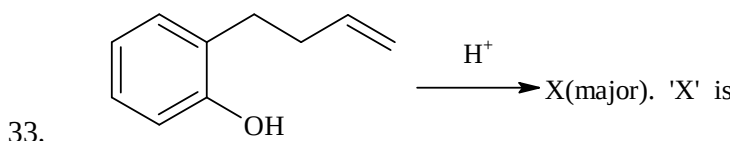
32. The correct order of ionic radii of Y^{3+} , La^{3+} , Eu^{3+} and Lu^{3+} is

A) $Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}$

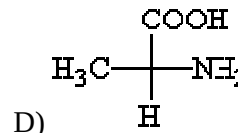
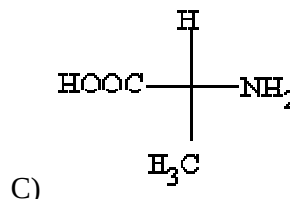
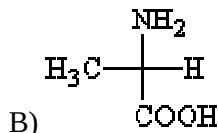
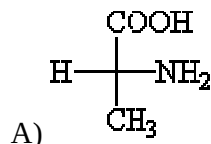
B) $La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}$

C) $Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$

D) $Lu^{3+} < La^{3+} < Eu^{3+} < Y^{3+}$



34. R - alanine is represented by



35. For the first order reaction $A_{(g)} \rightarrow 2B_{(g)} + C_{(g)}$, the initial pressure is $P_A = 90$ mm Hg, the pressure after 10 minutes is found to be 180 mm Hg. The rate constant of the reaction is

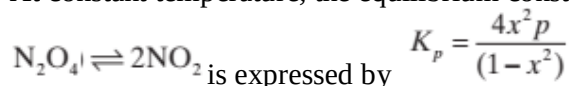
A) $1.15 \times 10^{-3} \text{ sec}^{-1}$

B) $2.3 \times 10^{-3} \text{ sec}^{-1}$

C) $3.45 \times 10^{-3} \text{ sec}^{-1}$

D) $6 \times 10^{-3} \text{ sec}^{-1}$

36. At constant temperature, the equilibrium constant (K) for the decomposition reaction



where p = pressure, x = extent of decomposition. Which one of the following statements is true?

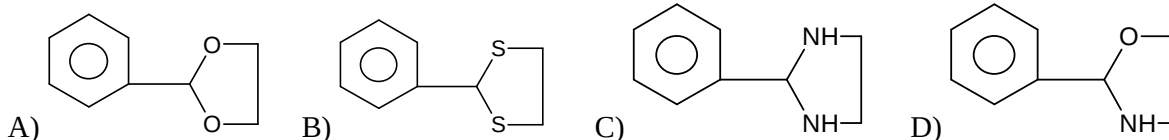
- A) K increases with increase of p B) K increases with increase of x
 C) K increases with decrease of x D) K remains constant with change in p and x
37. The non-stoichiometric compound $\text{Fe}_{0.94}\text{O}$ is formed when $x\%$ of Fe^{2+} ions are replaced by Fe^{3+} ions then x is
- A) 6 B) 12 C) 15 D) 18

38.

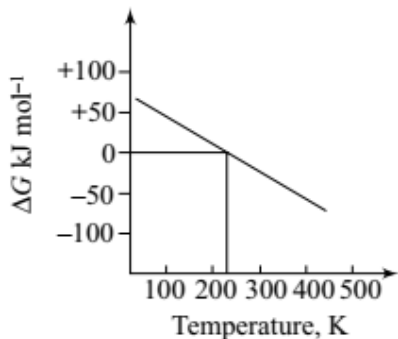
Gas	(Critical Temperature T_c (K))	Critical Pressure P_c (atm)
A	480	20
B	600	15
C	800	40
D	336	16

The gas having highest Vander Waals constant (b) is

- A) A B) B C) C D) D
39. Which of the following compound has most acidic hydrogen?

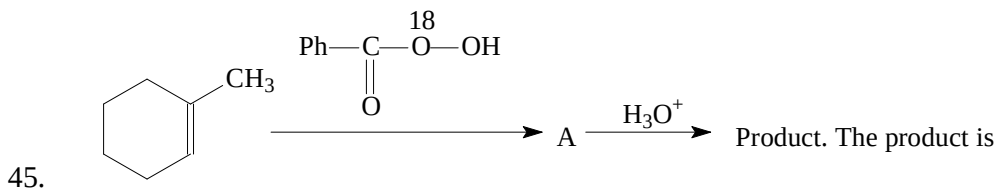


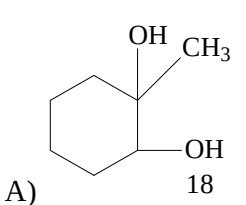
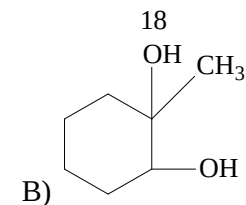
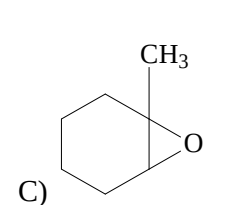
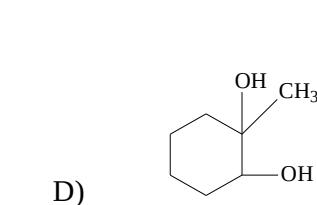
40. If the rate of reaction of [0.1 M] sodium cyanide with [0.1M] 1-bromoethane is 1.4×10^{-4} . What effect will an increase in NaCN concentration to [0.3M] and alkyl bromide concentration to [0.2M] have on the overall reaction rate?
- A) increase by 2 times B) increase by 3 times
 C) increase by 6 times D) increase by 1.5 times
41. What can be concluded about the values of ΔH and ΔS from this graph?



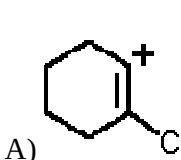
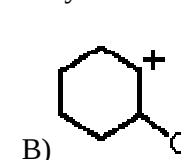
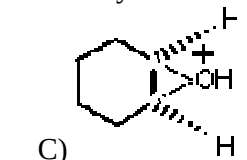
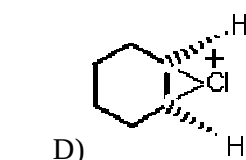
- A) $\Delta H > 0$, $\Delta S > 0$ B) $\Delta H > 0$, $\Delta S < 0$ C) $\Delta H < 0$, $\Delta S > 0$ D) $\Delta H < 0$, $\Delta S < 0$
42. 1L concentrated NaCl solution is electrolysed using Hg cathode. Calculate the pH of solution after passing 0.005 F charge to this solution. Assume volume of electrolyte remains constant during the electrolysis
- A) 12.7 B) 11.7 C) 5.3
 D) Will remain unchanged, i.e solution will have $\text{pH} = 7$

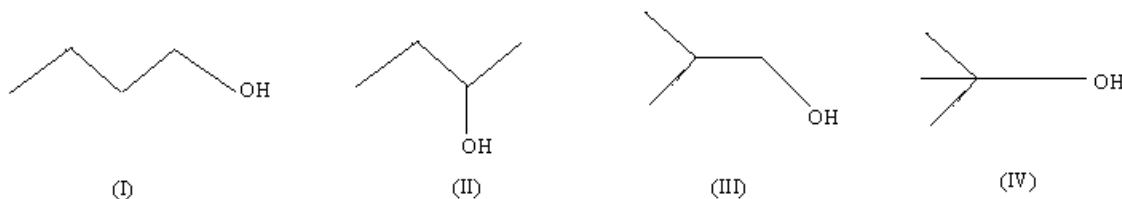
43. The protective power of a lyophilic colloid is measured by the property called
 A) Flocculation value B) Gold number
 C) Hardy Schulze rule D) Peptization
44. The $[H^+]$ of a solution containing 0.1M HA ($K_a = 12 \times 10^{-3}$) and 0.1M HB ($K_a = 24 \times 10^{-3}$) is
 A) 2×10^{-2} B) 4×10^{-2} C) 6×10^{-2} D) 10^{-2}



- A)  B)  C)  D) 

46. The order of increasing ionic radius of the following is:
 A) $K^+ < Mg^{2+} < Al^{3+}$ B) $K^+ < Mg^{2+} < Al^{3+}$
 C) $K^+ < Al^{3+} < Mg^{2+}$ D) $Al^{3+} < Mg^{2+} < K^+$
47. Which of the species shown below is the most stable form of the intermediate in the electrophilic addition of Cl_2 in water to cyclohexene to form a halohydrin?

- A)  B)  C)  D) 
48. Arrange the following alcohols in the increasing tendency of ester formation with benzoic acid in presence of H^+



- A) $II < IV < I < III$ B) $IV < II < III < I$ C) $I < II < III < IV$ D) $III < I < II < IV$

49. The main product in the Kolbe's electrolysis of $H_3C-CH(COO^-)-CH_3$ is



50. The orbital angular momentum corresponding to $n = 4$ and $m = -3$ is:

A) $\frac{h}{\sqrt{2}\pi}$

B) 0

C) $\frac{\sqrt{3}h}{\pi}$

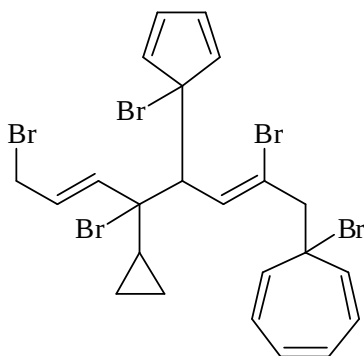
D) $\frac{\sqrt{6}h}{2\pi}$

SECTION – II (NUMERICAL VALUE ANSWER TYPE)

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51. When an oxide of manganese (A) is fused with KOH in the presence of an oxidizing agent and dissolved in H_2O , it gives a dark green solution of compound (B). Compound (B) disproportionates in neutral or acidic media to give purple compound (C). An alkaline solution of compound (C) oxidizes potassium iodide solution to a compound (D) and compound (A). In the conversion of C to D, the total no. of electrons involved in that stoichiometric equation is

52. The number moles of $AgNO_3$ required to produce a pale yellow precipitate from the following compound is

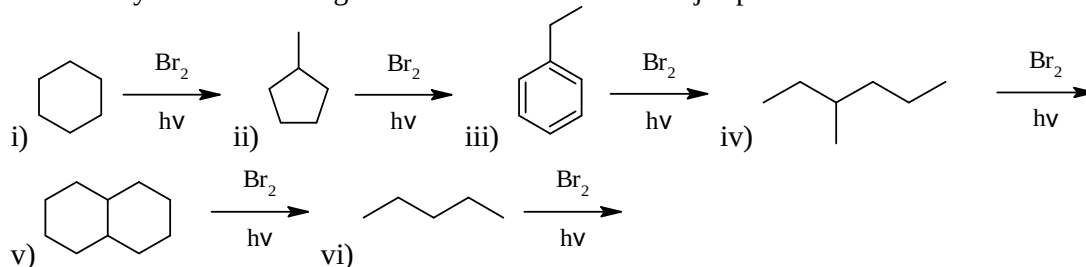


53. How many 3d- series elements have the standard $M^{2+}(aq)/M(s)$ potential with positive sign?

54. Two acid solutions A and B are titrated separately each with 25mL of 1N Na_2CO_3 solution. The volume of each acid used for titration are 10 ml and 40 ml respectively. Find out the volume ratio of acid B and A respectively, which should be mixed to prepare 1mL of 1N solution.

55. The number of compounds in the following which give oxygen on heating is HgO , MnO_2 , Pb_3O_4 , $KMnO_4$, $K_2Cr_2O_7$, Ag_2CO_3

56. In how many of the following reactions 2° halide is the major product?



57. How many of the following are correct about the stability of oxidation states of manganese?
- $Mn(II) > Mn(VII)$ - (acidic solution)
 - $Mn(II) < Mn(IV)$ - (alkaline solution)
 - $Mn(VI) > Mn(IV)$ - (acidic solution)
 - $Mn(VI) > Mn(VII)$ - (strong alkaline solution)
58. A solution of 0.01 M HCOOH ($K_a = 2 \times 10^{-4}$) and 0.01 M HCl has concentration of $HCOO^-$ equal to 2×10^{-n} . Then the value of n is
59. Bond order in carbonate ion is
60. No of halogen atoms present in the Halothane is

MATHEMATICS**Max.Marks:100****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

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61. $\int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1 + \sqrt[3]{x})} dx$ is equal to
- $\frac{3}{2}x^{2/3} + 6 \tan^{-1} x^{1/6} + C$
 - $\frac{3}{2}x^{2/3} - 6 \tan^{-1} x^{1/6} + C$
 - $\frac{-3}{2}x^{2/3} + 6 \tan^{-1} x^{1/6} + C$
 - None of these
62. If $\int f(x) \sin x \cos x dx = \frac{1}{2(b^2 - a^2)} \log[f(x)] + C$, then $f(x)$ is equal to
- $\frac{1}{a^2 \sin^2 x + b^2 \cos^2 x}$
 - $\frac{1}{a^2 \sin^2 x - b^2 \cos^2 x}$
 - $\frac{1}{a^2 \cos^2 x + b^2 \sin^2 x}$
 - $\frac{1}{a^2 \cos^2 x - b^2 \sin^2 x}$

63. $\int (x^{3a} + x^{2a} + x^a)(2x^{2a} + 3x^a + 6)^{1/a} dx$

A) $\frac{1}{6(a+1)}(2x^{3a} + 3x^{2a} + 6x^a)^{1+\frac{1}{a}} + C$ B) $\frac{1}{6(a+1)}(2x^{3a} + 3x^{2a} + 6x^a)^{1+\frac{1}{a}} + C$

C) $\frac{1}{3(a+1)}(2x^{3a} + 3x^{2a} + 6x^a)^{1+\frac{1}{a}} + C$ D) None of these

64. $\int \frac{x^2 - 1}{(x^2 + 1)\sqrt{1 + x^4}} dx = k \cos^{-1}\left(\frac{\sqrt{2}x}{x^2 + 1}\right) + C$, where $k =$

A) $\frac{1}{2}$ B) 2 C) $\frac{1}{\sqrt{2}}$ D) $\sqrt{2}$

65. $\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}} =$

A) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4} + C$ B) $\frac{3}{4}\left(\frac{x-1}{x+2}\right)^{1/4} + C$ C) $\frac{4}{3}\left(\frac{x+2}{x-1}\right)^{1/4} + C$ D) None of these

66. The integral $\int \left(1 + x - \frac{1}{x}\right) e^{x + \frac{1}{x}} dx$ is equal to

A) $(x-1)e^{x + \frac{1}{x} + C}$ B) $xe^{x + \frac{1}{x} + C}$ C) $(x+1)e^{x + \frac{1}{x} + C}$ D) $-xe^{x + \frac{1}{x} + C}$

67. The integral $\int \frac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx$ is equal to :

A) $\frac{-x^{10}}{2(x^5 + x^3 + 1)} + C$ B) $\frac{-x^5}{(x^5 + x^3 + 1)^2} + C$
 C) $\frac{-x^{10}}{2(x^5 + x^3 + 1)^2} + C$ D) $\frac{-x^5}{2(x^5 + x^3 + 1)^2} + C$

68. If $\int_0^{\pi} x f(\sin^3 x + \cos^2 x) dx =$

$k \int_0^{\pi/2} f(\sin^3 x + \cos^2 x) dx$, then $k =$

A) $\frac{\pi}{2}$ B) π C) 2π D) None of these

69. The value of the integral $\int_a^{a+\frac{\pi}{2}} (|\sin x| + |\cos x|) dx$ is
 A) $a\pi$ B) $2a\pi$ C) $\frac{a\pi}{2}$ D) Independent of a
70. $\int_0^{\pi} [\tan^{-1} x] dx$ is equal to (where $[\cdot]$ denotes greatest integer function)
 A) π B) $\tan 1$ C) $\pi + \tan 1$ D) $\pi - \tan 1$
71. $\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n}$ is equal to
 A) -1 B) e^{-1} C) 1 D) e
72. The area included between the curves $x^2 + y^2 = a^2$ and $\sqrt{|x|} + \sqrt{|y|} = \sqrt{a}$ ($a > 0$) is
 A) $\left(\pi + \frac{2}{3}\right)a^2$ B) $\left(\pi - \frac{2}{3}\right)a^2$ C) $\frac{2}{3}a^2$ D) $\frac{2\pi}{3}a^2$
73. The area of the region enclosed by the curves $y = x$, $x = e$, $y = 1/x$ and the positive x -axis is
 A) $5/2$ square units B) $1/2$ square units C) 1 square units D) $3/2$ square units
74. The area bounded between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$ and the straight line $y = 2$ is
 A) $20\sqrt{2}$ B) $\frac{10\sqrt{2}}{3}$ C) $\frac{20\sqrt{2}}{3}$ D) $10\sqrt{2}$
75. The area above x -axis, bounded by the line $x = 4$ and the curve $y = f(x)$, where $f(x) = x^2$, $0 \leq x \leq 1$ and $f(x) = \sqrt{x}$, $x \geq 1$, is
 A) 1 B) 2 C) 4 D) 5
76. The area of the portion of the circle $x^2 + y^2 = 1$, which lies inside the parabola $y^2 = 1 - x$, is
 A) $\frac{\pi}{2} - \frac{2}{3}$ B) $\frac{\pi}{2} + \frac{2}{3}$ C) $\frac{\pi}{2} + \frac{4}{3}$ D) $\frac{\pi}{2} - \frac{4}{3}$
77. Let f be integrable over $[0, a]$ for any real a . If we define $I_1 = \int_0^{\pi/2} \cos \theta f(\sin \theta + \cos^2 \theta) d\theta$ and $I_2 = \int_0^{\pi/2} \sin 2\theta f(\sin \theta + \cos^2 \theta) d\theta$, then
 A) $I_1 = I_2$ B) $I_1 = -I_2$ C) $I_1 = 2I_2$ D) $I_1 = -2I_2$
78. The order of the differential equation whose general solution is given by $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c}$, where c_1, c_2, c_3, c_4, c_5 are arbitrary constants, is
 A) 5 B) 4 C) 3 D) 2

79. The differential equation satisfied by $ax^2 + by^2 = 1$ is
- A) $xy \cdot \frac{d^2y}{dx^2} + x \cdot \left(\frac{dy}{dx}\right)^2 + y \cdot \frac{dy}{dx} = 0$ B) $xy \cdot \frac{d^2y}{dx^2} + x \cdot \left(\frac{dy}{dx}\right)^2 - y \cdot \frac{dy}{dx} = 0$
- C) $xy \cdot \frac{d^2y}{dx^2} - x \cdot \left(\frac{dy}{dx}\right)^2 + y \cdot \frac{dy}{dx} = 0$ D) None of these
80. The solution of the differential equation $x(y^2 e^{xy} + e^{x/y}) dy = y(e^{x/y} - y^2 e^{xy}) dx$ is
- A) $xy = \ln(e^{y/x} + c)$ B) $xy = \ln(e^{x/y} + c)$
- C) $\frac{y}{x} = \ln(e^{xy} + c)$ D) $\frac{x}{y} = \ln(e^{xy} + c)$

SECTION – II (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the numerical value has more than two decimal places, **Truncate/Round-off the value** of Two decimal places. **Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:
Marking scheme: +4 for correct answer, 0 in all other cases.

81. The value of the integral $\int_3^4 \frac{[x^2]}{[x^2 - 14x + 49] + [x^2]} dx$, where $[.]$ denotes the greatest integer function, is
82. The value of the integral $\int_0^\pi \frac{\sin 2kx}{\sin x} dx$ (where $k \in I$), is v then the value of $v + 2$ is _____
83. Let $f(x)$ be a continuous function such that the area bounded by the curve $y = f(x)$, x -axis and the lines $x = 0$ and $x = a$ is $\frac{a^2}{2} + \frac{a}{2} \sin a + \frac{\pi}{2} \cos a$, then $f\left(\frac{\pi}{2}\right) =$
84. The value of c for which the area of the figure bounded by the curve $y = 8x^2 - x^5$, the straight lines $x = 1$ and $x = c$ and the x -axis is equal to $16/3$, is
85. If $\int_0^x f(t) dt = x + \int_x^1 t f(t) dt$, then the value of $f(1)$ is
86. Suppose that $f''(x)$ is continuous for all x and $f(0) = f'(1) = 1$ If $\int_0^1 t f''(t) dt = 0$, then the value of $f(1)$ is
87. The order of the differential equation, of which $xy = ce^x + be^{-x} + x^2$ is a solution, is

88. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \cot^{-1} \left(r^2 + \frac{3}{4} \right) = \tan^{-1}(k)$ then $k =$ _____
89. Let $f(x) = [3 + 2 \cos x]$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$, where $[.]$ denotes the greatest integer function. The number of points of discontinuity of $f(x)$ is _____
90. If $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ for all $x \in R$, and $f(0) + f(2) = k f(1)$ then $k =$ _____

PAPER SETTERS – NLR

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CHEMISTRY	SRI C.S. MEHTA	9848194514